

Part A — Basics of Java Programming

Exercise 1

Write a Java program to print:

Welcome to Java Programming

Exercise 2

Write a Java program without the correct main() method signature. Compile and execute the program. Observe and explain the runtime error.

Exercise 3

Create two classes in a single Java file. Compile the program and observe the generated .class files.

Exercise 4

Write a Java program to demonstrate all primitive datatypes in Java.

Datatypes to include:

- byte
 - short
 - int
 - long
 - float
 - double
 - char
 - boolean
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Exercise 5

Write a Java program to display the minimum and maximum values of:

- Integer

- Byte
 - Short
 - Long
 - Float
 - Double
-

Part B — Type Conversion

Exercise 6

Write a Java program to demonstrate widening type conversion.

Convert:

- short to int
 - int to long
 - float to double
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Exercise 7

Write a Java program to demonstrate narrowing type conversion.

Convert:

- int to byte
- double to float
- long to int

Observe the data loss after conversion.

Part C — Operators and Input

Exercise 8

Write a Java program to calculate:

- total marks
- average marks

- percentage

for 5 subjects.

Exercise 9

Write a Java program to calculate employee salary details.

Requirements:

- Basic Salary
- HRA = 20%
- DA = 10%
- Tax = 5%

Display the salary slip.

Exercise 10

Write a Java program to swap two numbers:

1. Using temporary variable
 2. Without using temporary variable
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Part D — Classes and Objects

Exercise 11

Create a Student class with:

- studentId
- studentName
- department

Create multiple student objects and display their details.

Exercise 12

Create an Employee class with:

- empId
- empName
- salary

Add a method to display employee information.

Exercise 13

Create a Car class with:

- brand
- model
- price

Create objects for different cars and display details.

Part E — Constructors

Exercise 14

Write a Java program to initialize object values using constructors.

Exercise 15

Write a Java program to demonstrate constructor overloading.

Include:

- default constructor
 - single parameter constructor
 - multiple parameter constructor
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Exercise 16

Write a Java program using the this keyword to differentiate local and instance variables.

Part F — Encapsulation

Exercise 17

Create a BankAccount class with:

- private balance variable

Provide methods to:

- deposit amount
 - withdraw amount
 - check balance
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Exercise 18

Create a Student class with private variables and public getter/setter methods.

Part G — Inheritance

Exercise 19

Create:

- Animal class
- Dog class inheriting Animal

Demonstrate inheritance using methods.

Exercise 20

Create:

- Person
- Student
- Faculty

Demonstrate hierarchical inheritance.
